

Eigen

Tavian Cieslar

Eigen: a C++ linear algebra library

- Header only C++ library
- Includes matrices, vectors, dense and sparse solvers, and decompositions (LU, Cholesky, QR, SVD)
- Used by #include <Eigen/Dense>, nothing to link or build
- Repository: gitlab.com/libeigen/eigen

Who’s Involved

- Developers: maintained by volunteers, with ~55 contributors over the past year and ~660 over the project’s lifetime
- Affiliations: over 10 organizations based on email domains found in commits (AMD, Tenstorrent, Flatiron Institute, and various universities)
- Users: C++ developers who need fast linear algebra in their code

How the project functions

- Development on GitLab: issues and merge requests, along with discussion on Discord
- An average of ~25 merged merge requests and ~56 comments weekly over the past year
- Average commit: ~9 files, +222/-111 lines
- No CLA, contributors keep their copyright
- Full CI pipeline and test suite with over 1,400 registered tests

Accuracy, stability, and cost

From CONTRIBUTING.md: “For decompositions and solvers specifically, the bar is matching LAPACK on conditioning, pivoting strategy, and backward stability. Don't trade numerical robustness for speed in those code paths without explicit sign-off. ”

- !3469 BDCSVD: Keep the secular search from missing the root (stability)
- !3482 Tensor: Reduce FFT twiddle recurrence drift (accuracy)

What I ran

Example from the “Getting started” page in the documentation, added printing the determinant of the matrix

```cpp
#include <iostream>
#include <Eigen/Dense>

using Eigen::MatrixXd;

int main() {
  MatrixXd m(2, 2);
  m(0, 0) = 3;
  m(1, 0) = 2.5;
  m(0, 1) = -1;
  m(1, 1) = m(1, 0) + m(0, 1);
  std::cout << m << std::endl;
  std::cout << "Determinant = " << m.determinant() << std::endl;
}
```

```
  3  -1
2.5 1.5
Determinant = 7
```

Built and ran all nine parts of the dense LU test, all passed

```
100% tests passed out of 9

Label Time Summary:
Official      =   9.19 sec*proc (9 tests)
smoketest     =   0.00 sec*proc (1 test)

Total Test time (real) =   9.25 sec
```

My Question

How does Eigen’s accuracy and speed compare to LAPACK, and how could that be measured/compared in a meaningful way?

Proposed group experiment

- Question: how do Eigen’s solvers trade accuracy for cost on ill-conditioned problems?
- Setup: solve Ax=b with various solvers from Eigen
- Figure 1: x-axis condition number, y-axis relative error, and a line for each solver
- Figure 2: x-axis matrix size, y-axis time, a line for each of the same solvers